

CellFoundry

Adipocyte and Stem Cell Instance Segmentation

IN COLLABORATION WITH **Institut für Medizintechnik Schweinfurt (IMES) – THWS:** Research partner, provider of microscopy imaging data and biomedical domain expertise.

Why it matters

This project serves two distinct motivations, corresponding to the two species of tissue studied. **Human tissue** supports biomedical research (obesity, type 2 diabetes, metabolic syndrome) and cell culture for drug testing ahead of clinical trials, with a partner institute in Hannover. **Porcine tissue** supports a cultured-meat application: cultured (lab-grown) meat is produced by growing animal cells in vitro, and since muscle tissue alone does not reproduce the flavor of conventionally farmed meat, fat cells (adipocytes) are deliberately included in the culture to contribute flavor.

The cultures are mixed populations of adipocytes and mesenchymal stem cells (bone-marrow derived, capable of differentiating into bone cells or adipocytes); the exact proportion is an estimate, not a fixed target. Manual annotation of microscopy images is not scalable to the volumes required. As differentiation progresses, adipocytes develop larger, connected lipid-droplet clusters, particularly in human-derived cultures; individual cell boundaries become increasingly difficult to identify, and a confirmed failure mode of IMES's current tools is incorrect segmentation of overlapping or intersecting cells. One useful visual cue: empty regions of an image tend to correspond to stem cells, while dotted regions (plausibly lipid droplets) correspond to adipose tissue – a pattern direct inspection of the project's own data supports, though not yet confirmed systematically.

The project

Can automated instance segmentation reliably identify individual adipocytes and stem cells, including distinguishing between the two cell states, in both brightfield and phase-contrast microscopy, across both porcine and human tissue, at a quality that outperforms the tools IMES currently uses?

IMES's current tools are **Cellpose-SAM** and **Dinocell** (Dinocell has reportedly outperformed Cellpose-SAM in IMES's own testing); these are the baseline to beat, not a prescribed toolset. Candidate approaches are open, including other SAM-family foundation models and any technique that can reasonably outperform the baseline.

Preprocessing is necessary: illumination correction, contrast enhancement (e.g. CLAHE), and intensity normalisation before segmentation; careful data augmentation (geometric transforms, intensity jitter, elastic deformations) at the fine-tuning stage.

Recommended sequence, metrics first: (1) determine the metrics IMES already uses and what IMES wants; (2) reproduce IMES's own prior Cellpose-SAM/Dinocell results as a validation checkpoint; (3) segment porcine cells first, the easier case; (4) segment human cells, partial/high-confidence-only segmentation acceptable; (5) compute cell count; (6) compute cell size and morphology, including the **width-to-length ratio** of each annotated cell; (7) apply pseudo-labelling to unannotated IMES images.

Evaluation metrics are not yet decided, and choosing them is itself part of the project's work – candidates include IoU/Dice, count accuracy, and size-distribution accuracy. Once chosen, Cellpose-SAM and Dinocell's baseline performance on those metrics must be established before any candidate model is compared against them.

Deliverable: a documented, reproducible evaluation pipeline; a method for distinguishing adipocyte and stem-cell states; and a recommended approach outperforming the Cellpose-SAM/Dinocell baseline.

Methods & starting points

- **Cellpose-SAM** – current IMES baseline tool.
- **Dinocell** – current IMES baseline tool, reportedly outperforming Cellpose-SAM.
- Candidates discussed so far: MicroSAM, CellSAM, OwlV2, SOwIV2, Omnipose, MedSAM, and optionally SAM3-Adapter and Medical SAM3 – list may grow, not a mandated scope.

WHAT WOULD SUPPORT OUR ANALYSIS

Reliably segment individual cells that are sufficiently distinguishable from one another. Within larger clusters and connected cell areas, identify individual cells or cell structures as far as possible, so that at least an approximate cell count can be obtained.

Scope spans:

- Two species (porcine, easier; human, harder)
- Two modalities (brightfield, phase-contrast; neither prioritized)
- Two cell states (adipocyte, stem cell; distinguished visually by morphology, not by type labels, which do not exist in the current annotations)